MART-NILM: A Multi-Appliance Relational Temporal Network for Joint Power and State Estimation in Non-Intrusive Load Monitoring

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***Abstract*—**

***Index Terms*—non-intrusive load monitoring, energy disaggregation, multi-task learning, power and state prediction, household appliances**

I. Introduction

Reducing residential carbon footprints and curbing greenhouse gas emissions depend on actionable, fine-grained visibility into household electricity consumption [1]. Although direct sub-metering can provide itemized feedback to guide conservation, installing dedicated hardware sensors on every appliance is intrusive, expensive, and difficult to scale [2]. Non-intrusive load monitoring (NILM) addresses this bottleneck by estimating the power and the on–off state of individual appliances from a single aggregate meter, without a sensor on each appliance. A recent systematic review groups more than thirty years of work into deterministic low-frequency methods, probabilistic and optimization models, and deep networks, and finds that reported accuracy depends as much on the sensing setup and the evaluation protocol as on the model [3]. In homes, deep learning is now the usual approach, while data requirements, real-time use, and interpretability remain open [4].

In conventional single-appliance NILM, a separate network is trained for each appliance. While single-appliance models dominate the existing literature, their linear scaling in computational footprint and memory storage limits real-world smart-meter deployment, leading recent reviews to highlight multi-appliance NILM as a crucial future research direction [3], [4]. By mapping one aggregate window to several loads simultaneously within a single network, multi-appliance NILM provides constant inference overhead and preserves joint operational context [5], [6]. However, it remains far less explored due to optimization challenges such as gradient conflict and cross-appliance scale imbalance [7]. Furthermore, existing multi-appliance designs typically adopt a shared feature extractor that merely feeds into separate output heads, whereas the truly beneficial form of joint disaggregation is capturing explicit relational dependencies among appliances.

A joint network is not, by itself, evidence about an unseen home. The measured aggregate is

$$x_t = \sum_{i=1}^{M} P_{t,i} + b_t + \epsilon_t, \tag{1}$$

where t is time, $i = 1, \ldots, M$ indexes the target appliances, M is their number, $P_{t,i}$ is the power of appliance i, b_t is the unmonitored background, and ϵ_t collects measurement error and timing inconsistency. The background includes every load outside the target set. A model can treat patterns in b_t as evidence for a target that merely co-occurs with them [8]. This term is not a small disturbance: it can be far larger than the roughly 80 W demand of a fridge. Two problems follow. A rare on event may be seen only against the backgrounds of the training houses, so the network can treat that background as evidence for the appliance. Joint training creates a second problem. For one appliance, power is a regression loss and the on–off state is a classification loss. The two losses do not share a scale, and the scale is different for each appliance. One ratio taken over all appliances would multiply every classification loss by the same number, so a large regression error on the microwave would change the weight of the fridge's classification error.

We propose MART-NILM, a multi-appliance relational temporal network that reads the aggregate and estimates the five appliances together. During training, a real on event is

placed against different levels of the remaining household load. The recorded meter is left unchanged at test time. For each appliance, the regression loss and the classification loss are placed on one scale and added. The training objective is the sum of these appliance losses. Training, validation, and test use different houses.

The contributions are threefold. First, training, validation, and test use different houses, the test meter keeps its recorded residual, and five appliances are scored together on UK-DALE and REFIT. Second, training shows rare on events against a range of the remaining household load, and this step adds nothing at test time. Third, the objective is the sum over appliances of a regression loss and a classification loss. The classification loss of each appliance is scaled by that appliance's own ratio of the two losses.

II. Related Work

Multi-appliance NILM replaces a collection of appliance-specific models with one network that estimates several loads jointly. UNet-NILM, for example, estimates the on–off state and power of several appliances with a single model [5]. Table I compares recent multi-appliance methods by evaluation scenario, data split, architecture, and principal contribution.

Cross-house evaluation is present in several studies. MART-NILM differs in that the validation houses are also held out and the test retains the recorded aggregate, including unmonitored loads, in two datasets.

III. Methodology

A. Problem Formulation

Let x_t denote aggregate power and $P_{t,i}$ the power of target appliance i, as in (1). From a standardized aggregate window $\mathbf{x} \in \mathbb{R}^T$, the model jointly estimates the standardized power and on-state probability of $M = 5$ appliances,

$$\hat{\mathbf{Y}} \in \mathbb{R}^{T\times M}, \qquad \hat{\mathbf{Z}} \in [0,1]^{T\times M}. \tag{2}$$

The input contains no appliance measurements or hand-crafted channels. The unobserved remainder is

$$x_t - \sum_{i=1}^{M} P_{t,i} = b_t + \epsilon_t. \tag{3}$$

Hence the network must distinguish appliance activity from both other targets and unmonitored household demand.

B. Model Architecture

MART-NILM follows the shared–private design in Fig. 1. A shared encoder extracts evidence once from the aggregate; appliance-specific decoders then separate this evidence into five streams; finally, relation attention allows the streams to exchange information before joint power and state prediction. Temporal resolution is preserved throughout.

Shared aggregate encoder: The encoder combines a multi-branch convolutional neural network (CNN) stem [14] with residual temporal convolution. Parallel kernels capture switching patterns at complementary local scales. To prevent the multi-scale CNN from overfitting to domain-specific baselines, instance–batch normalization (IBN) [15] divides shallow channels equally between instance normalization (IN) and batch normalization (BN). This allows the convolutional kernels to extract house-invariant switching transients via instance normalization, while preserving the physical wattage scale through batch normalization for cross-house generalization. The resulting sequence is processed by five dilated temporal convolutional network (TCN) blocks [16]. Writing $\mathcal{T}_\ell$ for the 1D convolution, batch normalization, ReLU activation, and dropout in block ℓ,

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \mathcal{T}_\ell\Big(\mathbf{H}^{(\ell)}; d_\ell\Big), \qquad d_\ell \in \{1,2,4,8,16\}. \tag{4}$$

This produces the shared representation $\mathbf{H}$ without temporal downsampling, preserving the alignment between input transitions and appliance outputs.

Appliance decoding and relation modelling: Each decoder applies a channel gate [17] and a local residual transformation to obtain an appliance-specific representation $\mathbf{F}_i$. These private streams prevent all appliances from being forced into the same representation. To model simultaneous operation, scaled dot-product attention [18] is evaluated across appliances at each time t, rather than across time:

$$\mathbf{A}_t = \mathrm{softmax}\left(\frac{\mathbf{Q}_t\mathbf{K}_t^{\mathsf{T}}}{\sqrt{d}}\right), \qquad \mathbf{M}_t = \mathbf{A}_t\mathbf{V}_t. \tag{5}$$

Here $\mathbf{Q}_t$, $\mathbf{K}_t$, and $\mathbf{V}_t$ are learned projections of the stacked appliance representations, and d is their projection width. A gated residual connection adds the relation message $\mathbf{M}_t$ to each private stream, so cross-appliance context supplements rather than replaces appliance-specific evidence.

Joint power and state prediction: For each appliance, two time-shared fully connected (FC) heads produce a raw standardized power $\hat{y}_i^{\mathrm{raw}}(t)$ and a state logit $a_i(t)$. The on-state probability $\hat{p}_i(t) = \sigma(a_i(t))$ is both the state prediction and a differentiable gate on the power branch [7], [19]:

$$\hat{y}_i(t) = \hat{p}_i(t)\hat{y}_i^{\mathrm{raw}}(t) + [1-\hat{p}_i(t)]y_i^{\mathrm{off}}, \tag{6}$$

where $y_i^{\mathrm{off}} = -\mu_i/\sigma_i$ is zero watts on the standardized scale. Thus an on probability near one preserves the regressed waveform, whereas a probability near zero moves the estimate towards zero watts. The final prediction in physical units is

$$\hat{P}_i(t) = [\sigma_i\hat{y}_i(t) + \mu_i]_+, \qquad [u]_+ = \max(u,0). \tag{7}$$

The state head therefore provides an explicit operating-state prior while the power head retains the appliance waveform.

C. Loss Function

The model is trained jointly over all appliances, as in multi-task NILM [5], [7], [11]. Let $b = 1,\ldots,B$ index windows in a mini-batch, $t = 1,\ldots,T$ index samples, and $i = 1,\ldots,M$ index appliances. The standardized target and predicted powers are $y_{b,t,i}$ and $\hat{y}_{b,t,i}$; $z_{b,t,i} \in \{0,1\}$ is the on–off label; and

TABLE I

MULTI-APPLIANCE STUDIES. THE SPLIT COLUMN STATES HOW TRAINING, VALIDATION, AND TEST ARE DIVIDED. AN ENTRY $a \rightarrow b$ MEANS TRAINING ON HOUSES a AND TESTING ON HOUSES b. "VAL. IN TRAIN" MEANS VALIDATION USES PART OF A TRAINING HOUSE.

Study	Scenario	Train / val. / test	Architecture	Contribution
UNet-NILM [5]	Inside one house	One house: UK-DALE 1, Jan.–Mar. 2015	1-D U-Net	Joint state and power from a sum of the monitored appliances
SAMNet [9]	Cross-house	Train→test: UK-DALE $1, 5 \rightarrow 2$; REFIT $2\text{–}6 \rightarrow 1$	Shared experts and self-attention	Joint state and power on an unseen house
Conv-NILM-Net [10]	Inside each house	10-fold inside each REDD building	Causal temporal convolution	One network separates several appliances at once
Virtsionis et al. [11]	One training house	3 tests, all trained on UK-DALE 1: house 1; house 2; REFIT 2, 5	Variational CNN, one head per appliance	One model estimates every target appliance
Li et al. [6]	Cross-house	4 cases: pretrain REFIT 11, then fine-tune and test the target houses	CNN with transferable heads	One model transferred across houses and datasets
Tanoni et al. [12]	Cross-house	Val. in train. Test: UK-DALE 2; REFIT 4, 9, 15	Convolutional recurrent network	Appliance states from coarse labels
MATNilm [7]	Cross-house	S1: REDD $2\text{–}6 \rightarrow 1$, val. in 2, 3. S2: 1-day train and val., unseen test. S3: S2 plus profiles	Convolutional encoder and attention	Joint state and power from little labeled data
Dash and Sahoo [13]	Inside and cross-house	Train REFIT 2. Three 1-week tests: house 2, 9, and 20	Dilated causal convolution and attention	One model detects several appliance states
Liu and Fan [8]	Cross-house	Separate datasets. Val. is the tail of each train house. Test: REDD 1, UK-DALE 2, REFIT 2	Two-stage mixture of experts	Estimates that stay stable when the unmonitored load changes

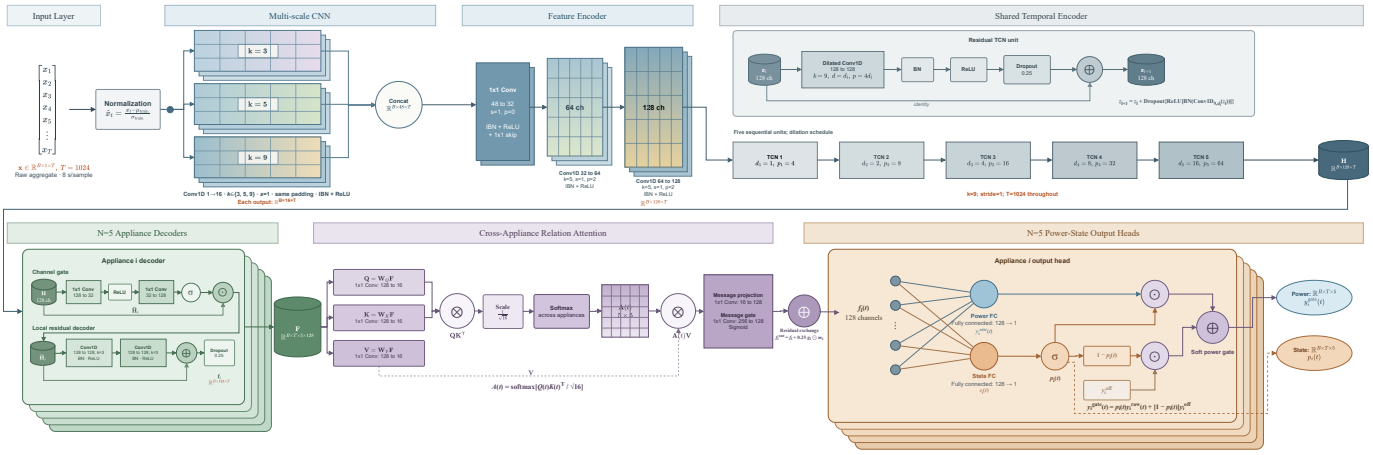


Fig. 1. Architecture of MART-NILM. The aggregate input is processed by a multi-scale convolutional stem and a shared temporal encoder. Appliance-specific decoders form five feature streams, cross-appliance attention exchanges information among them, and separate power and state heads jointly produce gated power estimates and on–off probabilities.

$\hat{p}_{b,t,i} \in (0,1)$ is the predicted on probability. The objective separates power regression from state classification:

$$\mathcal{L} = \underbrace{\sum_{i=1}^{M} \mathcal{L}_{\text{power},i}}_{\mathcal{L}_{\text{power}}} + \lambda \underbrace{\sum_{i=1}^{M} r_i \mathcal{L}_{\text{state},i}}_{\mathcal{L}_{\text{state}}}, \qquad \lambda = 0.8. \tag{8}$$

Hence every appliance contributes one power loss and one state loss, and the five appliance losses are summed to train a single network.

Power loss: For appliance i, let $e_{b,t,i} = \hat{y}_{b,t,i} - y_{b,t,i}$, $N_i^{\text{on}} = \sum_{b,t} z_{b,t,i}$, and $N_i^{\text{off}} = BT - N_i^{\text{on}}$. The overall, on-state, and off-state mean squared errors are

$$\begin{aligned} \text{MSE}_i &= \frac{1}{BT} \sum_{b,t} e_{b,t,i}^2, \\ \text{MSE}_i^{\text{on}} &= \frac{\sum_{b,t} z_{b,t,i} e_{b,t,i}^2}{\max(N_i^{\text{on}}, 1)}, \\ \text{MSE}_i^{\text{off}} &= \frac{\sum_{b,t} (1 - z_{b,t,i}) e_{b,t,i}^2}{\max(N_i^{\text{off}}, 1)}. \end{aligned} \tag{9}$$

The conditional terms keep errors during the sparse on state visible instead of allowing the many off samples to dominate the regression objective.

To preserve switching behaviour, define $\Delta y_{b,t,i} = y_{b,t,i} - y_{b,t-1,i}$ and $m_{b,t,i} = \max(z_{b,t,i}, z_{b,t-1,i})$. With $N_i^{\Delta} =$

$\sum_{b,t=2}^{T} m_{b,t,i}$,

$$\mathrm{MSE}_i^{\Delta} = \frac{\sum_{b,\,t=2}^{T} m_{b,t,i}(\Delta\hat{y}_{b,t,i} - \Delta y_{b,t,i})^2}{\max(N_i^{\Delta}, 1)}. \tag{10}$$

Window-level energy is measured in watts using $P_{b,t,i} = [\sigma_i y_{b,t,i} + \mu_i]_+$ and $\hat{P}_{b,t,i} = [\sigma_i \hat{y}_{b,t,i} + \mu_i]_+$:

$$\mathrm{RE}_i = \frac{1}{B}\sum_{b=1}^{B} \frac{\left|\sum_{t=1}^{T}(\hat{P}_{b,t,i} - P_{b,t,i})\right|}{\sum_{t=1}^{T} P_{b,t,i} + TP_0}, \quad P_0 = 10 \text{ W}. \tag{11}$$

The per-appliance power loss is therefore

$$\begin{aligned}\mathcal{L}_{\mathrm{power},i} = \mathrm{MSE}_i + \mathrm{MSE}_i^{\mathrm{on}} + 0.5\,\mathrm{MSE}_i^{\mathrm{off}} \\ + 0.15\,\mathrm{MSE}_i^{\Delta} + 0.25\,\mathrm{RE}_i.\end{aligned} \tag{12}$$

State loss: Let π_i denote the proportion of on samples for appliance i in the training set. Its positive-class weight is $w_i = \min\{(1-\pi_i)/\pi_i, 12\}$. The weighted binary cross-entropy is

$$\begin{aligned}\mathrm{BCE}_i = -\frac{1}{BT}\sum_{b,t}\big[w_i z_{b,t,i}\log\hat{p}_{b,t,i} \\ + (1 - z_{b,t,i})\log(1-\hat{p}_{b,t,i})\big].\end{aligned} \tag{13}$$

An additional false-positive term penalizes high on probabilities when the appliance is off:

$$\mathcal{L}_{\mathrm{fp},i} = \frac{\sum_{b,t}(1 - z_{b,t,i})\hat{p}_{b,t,i}^2}{\max(N_i^{\mathrm{off}}, 1)}, \qquad \mathcal{L}_{\mathrm{state},i} = \mathrm{BCE}_i + \mathcal{L}_{\mathrm{fp},i}. \tag{14}$$

Appliance-wise task balance: Because MSE and BCE have different numerical scales, the state loss of each appliance is matched to its own power loss. Let

$$\begin{aligned} r &= \mathrm{sg}\left(\frac{\mathcal{L}_{\mathrm{power}}}{\sum_j \mathcal{L}_{\mathrm{state},j} + \varepsilon}\right), \\ q_i &= \frac{\mathcal{L}_{\mathrm{power},i}}{\mathcal{L}_{\mathrm{state},i} + \varepsilon}, \qquad r_i = \mathrm{sg}\Big[\mathrm{clip}\Big(q_i, \frac{r}{3}, 3r\Big)\Big]. \end{aligned} \tag{15}$$

Here sg stops gradients through the weights and $\varepsilon = 10^{-8}$ prevents division by zero. Substitution of r_i into Eq. (8) yields the complete multi-appliance objective while preventing one appliance from setting the classification scale of all others.

D. Training Strategy

Training alternates between recorded and recomposed aggregate windows with equal probability. The recorded windows retain the natural joint distribution of household loads. The recomposed windows reduce dependence on appliance combinations and residual backgrounds observed in a particular house, following the use of synthetic aggregates in NILM [8], [20]. From each recorded training sequence, the non-target residual is

$$R_t = \left[x_t - \sum_{i=1}^{M} P_{t,i}\right]_+. \tag{16}$$

TABLE II
WHOLE-HOUSE SPLIT.

Split	Houses	Block
Training	UK-DALE house 1; REFIT houses 3, 5, 9, 11	5 weeks
Validation	UK-DALE house 5; REFIT house 2	5 weeks
Test	UK-DALE house 2; REFIT house 20	5 weeks

For a recomposed window, each appliance trace and the residual are drawn from independently selected training windows:

$$\tilde{x}_t = R_t^{(k_0)} + \sum_{i=1}^{M} P_{t,i}^{(k_i)}, \qquad \tilde{P}_{t,i} = P_{t,i}^{(k_i)}, \quad \tilde{z}_{t,i} = z_{t,i}^{(k_i)}, \tag{17}$$

where $k_0, k_1, \ldots, k_M$ denote independent window indices. The additive construction therefore preserves the power and state labels of every selected appliance.

To prevent sparse events from disappearing from the recomposed set, one focal appliance $i^\star$ is selected uniformly for each synthetic window and its trace is drawn from windows containing at least one on sample. The residual window is selected by first sampling uniformly from the non-empty median-power intervals $[0, 100)$, $[100, 200)$, $[200, 400)$, $[400, 800)$, and $[800, \infty)$ W, and then sampling a window within that interval. The remaining appliance traces are sampled without this constraint. Recomposition is used only for training; validation and test retain the recorded aggregate and its aligned labels.

IV. Experimental Setup

A. Dataset

Two public datasets are used: UK-DALE [21] and REFIT [22]. The target appliances are the kettle, fridge, dishwasher, washing machine, and microwave, and all signals are resampled at 8 s.

The experiment uses a whole-house split. Each house belongs to only one of training, validation, and test, as listed in Table II. Every house contributes one continuous 5-week block. The held-out blocks are centred within the previously established eight-week recording intervals and are selected without reference to appliance events or model performance. Normalization statistics are estimated from the training houses only.

B. Data Preprocessing

Aggregate and appliance channels are independently mean-resampled onto an 8 s grid. The primary experiments retain the recorded timing; Appendix A reports a separate timing-sensitivity audit that was not used for model selection.

Let μ_x, σ_x and μ_i, σ_i be the aggregate and appliance statistics estimated from the training houses. The input and target powers are standardized as

$$\bar{x}_{n,t} = \frac{x_{n,t} - \mu_x}{\sigma_x}, \qquad y_{n,t,i} = \frac{P_{n,t,i} - \mu_i}{\sigma_i}. \tag{18}$$

State labels are obtained before windowing from $z_{t,i} = \mathbb{I}[P_{t,i} \geq \tau_i]$. Off gaps no longer than G_i are joined to the surrounding on period, and on periods shorter than D_i are

TABLE III
ON-OFF LABELING PARAMETERS.

Appliance	τ (W)	G (s)	D (s)		Spike
			UK-DALE	REFIT	
Kettle	200	0	12	12	No
Fridge	50	12	60	60	Yes
Dishwasher	50	1800	960	960	Yes
Washing machine	20	960	960	960	Yes
Microwave	200	30	12	6	No

TABLE IV
TRAINING HYPERPARAMETERS.

Model	Window	Stride	Batch	LR	Epochs
UNet-NILM [5]	100	1	256	10^{-3}	50
MAT-Conv [7]	864	64	64	10^{-3}	200
MART-NILM	1024	512	64	10^{-4}	150

discarded. Table III gives the corresponding values for each dataset, with a 30 s fridge gap used in UK-DALE house 5. Isolated spikes are removed only for the appliances marked in the table.

C. Experimental Design

MART-NILM is compared with UNet-NILM [5] and MAT-Conv [7]. Each baseline keeps the hyperparameters of its own paper. Table IV lists their settings. All three are trained with Adam. UNet-NILM runs for the full 50 epochs. MAT-Conv stops after 30 epochs without improvement. MART-NILM uses weight decay 10^{-4} and unit-norm gradient clipping. Its learning rate is reduced by a factor of 0.5 after 12 epochs without validation improvement, to a minimum of 10^{-5}. Training is limited to 150 epochs; early stopping is disabled before epoch 100 and thereafter uses a patience of 55 epochs. A 1024-sample window covers about 136.5 min.

Model selection and state calibration use only the validation houses. The retained checkpoint minimizes

$$S_{\mathrm{val}} = \mathrm{MAE}_{\mathrm{val}}^{\mathrm{norm}} + 1 - \mathrm{AP}_{\mathrm{val}}^{\mathrm{macro}}, \tag{19}$$

where average precision (AP) is the area under the precision–recall curve computed from the continuous state probabilities. After checkpoint selection, an appliance-specific threshold is chosen from $\{0.05, 0.06, \ldots, 0.98\}$ to maximize validation F1. For the microwave, predicted on periods shorter than 32 s are removed and off gaps no longer than 24 s are joined. The resulting binary state $\hat{z}_{t,i}$ is also applied to the power output, $\hat{P}_{t,i}^{\mathrm{eval}} = \hat{z}_{t,i}\hat{P}_{t,i}$. The fitted thresholds and temporal rules are then frozen and applied unchanged to both test houses.

D. Evaluation Metrics

Each appliance is evaluated on power and on the on–off state. For an evaluated record of length T, P_t is the recorded power in watts and $\hat{P}_t$ is the predicted power. The label z_t equals 1 when the appliance is recorded on and 0 when it is off. $\hat{z}_t$ is the predicted label. The Average column in Table V is the mean of the five appliance scores. Here T is the length of the record, not the training window of Section III-A.

For an appliance-level score $q_{h,i}$ of appliance i in house h, we first form the appliance average within that house and then give every house equal weight,

$$\bar{q}_h = \frac{1}{M}\sum_{i=1}^{M} q_{h,i}, \qquad \bar{q}_{\mathcal{H}} = \frac{1}{|\mathcal{H}|}\sum_{h\in\mathcal{H}} \bar{q}_h. \tag{20}$$

For F1, $\bar{q}_h$ is the macro-F1 over the five appliances. Model and checkpoint selection use $\mathcal{H} = \{\text{UK-DALE 5}, \text{REFIT 2}\}$, the two validation houses. This separation prevents repeated design choices from biasing the final evaluation [23]. UK-DALE 2 and REFIT 20 are excluded from selection and are reported separately as cross-house test cases. Their equal-house mean is used in the architecture-ablation summary only after every checkpoint has been frozen; it never replaces the two house-specific results in Table V.

1) Power: The same absolute error is reported in three ways. MAE uses every sample. ON MAE uses samples with $z_t = 1$. OFF MAE uses samples with $z_t = 0$.

$$\mathrm{MAE} = \frac{1}{T}\sum_{t=1}^{T}\left|P_t - \hat{P}_t\right|. \tag{21}$$

SAE divides the record into consecutive blocks of $L = 1200$ samples, or 160 min at 8 s. The index of a block is k, and $K = \lfloor T/L \rfloor$ is the number of complete blocks. A trailing block shorter than L is omitted. SAE is the mean absolute error of the average power in each block,

$$\mathrm{SAE} = \frac{1}{K}\sum_{k=1}^{K}\left|\frac{1}{L}\sum_{t=(k-1)L+1}^{kL}\left(P_t - \hat{P}_t\right)\right|. \tag{22}$$

2) On-Off State: The recorded label z_t follows the rules in Table III, and $\hat{z}_t$ is the predicted binary state. For each appliance, TP (true positives) counts samples correctly identified as on ($z_t = 1, \hat{z}_t = 1$), FP (false positives) counts inactive samples falsely detected as on ($z_t = 0, \hat{z}_t = 1$), and FN (false negatives) counts active samples missed as off ($z_t = 1, \hat{z}_t = 0$):

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad \mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{23}$$

$$\mathrm{F1} = \frac{2\cdot\mathrm{Precision}\cdot\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}}. \tag{24}$$

Precision is the fraction of predicted on samples that are truly on, and recall is the fraction of actual on samples detected. F1 is their harmonic mean; true negatives (TN, $z_t = 0, \hat{z}_t = 0$) are excluded to prevent long idle background periods from artificially inflating the score.

V. RESULTS

A. Power Disaggregation and State Detection

Table V compares each model on UK-DALE and REFIT. DW, WM, and MW are the dishwasher, washing machine, and microwave. Each entry is the mean and standard deviation

over three seeds; entries with ± 0.0 or ± 0.00 indicate values below the reported display precision. The scores are defined in Section IV-D. Lower power error is better. Higher precision, recall, and F1 are better.

B. Computational Cost

Table VI reports the number of trainable parameters, checkpoint size, and training time of each model on one GPU. Checkpoint size is measured from the saved `best.pt` file using $1\,\mathrm{MB} = 1024^2$ bytes.

C. Ablation Study

Architecture and training effects are evaluated separately. Every architecture variant in Table VII remains one joint five-appliance network and retains five power and five state outputs. A removed module is replaced only by an identity mapping or the pointwise projection required to preserve tensor dimensions. Each row therefore removes one complete block shown in Fig. 1, while all other settings remain fixed. Checkpoints are selected on the validation houses. The table then reports frozen evaluation on UK-DALE house 2 and REFIT house 20: each metric is calculated separately for each house and the two values are averaged with equal weight using (20).

The multi-scale CNN is reduced to its central kernel while preserving its width and residual path. Removing the feature encoder retains only a pointwise channel projection, and removing the temporal encoder uses an identity mapping. The appliance-decoder ablation sends the shared temporal representation directly to five independent power/state readouts. The final variant retains these appliance-specific readouts but removes information exchange between them. These replacements preserve the joint prediction task and isolate the contribution of each top-level module.

The shared temporal encoder gives the largest contribution: its removal increases MAE by 1.44 W and ON MAE by 45.13 W, while macro-F1 falls from 0.796 to 0.710. Removing the appliance-specific decoders increases ON MAE by 22.90 W and reduces macro-F1 by 0.013. The feature encoder provides a smaller but consistent benefit to ON-state reconstruction and detection. The multi-scale CNN has little effect on the aggregate averages, but its benefit is concentrated on the sparse microwave task: on validation UK-DALE 5 its removal increases microwave ON MAE from 589.49 to 696.45 W, and on REFIT 20 microwave F1 falls from 0.549 to 0.512. Relation attention also exhibits a task trade-off. Removing it lowers average ON MAE, but reduces macro-F1 and lowers REFIT 20 microwave F1 from 0.549 to 0.496. Waveform inspection agrees with the numerical results: removing temporal encoding produces prolonged false fridge activations and poorly aligned microwave plateaus, while removing the appliance decoders increases fragmentation and amplitude error.

Table VIII freezes the complete architecture and examines the two proposed training strategies. Standard joint training uses recorded windows and a fixed power–state weight. The next two rows add focal-event mixing and appliance-wise balancing independently, and complete MART-NILM combines both strategies.

Table VIII supports both strategies, but also exposes their different effects. Focal-event mixing improves macro-F1 by 0.044 and ON MAE by 56.19 W relative to standard training, while its aggregate MAE is 0.52 W higher. It should therefore be interpreted as a coverage intervention rather than a generic regularizer. Appliance-wise balancing improves all four reported measures when used alone. Their combination gives the best MAE, SAE, and macro-F1, raising macro-F1 by 0.082 over standard joint training.

Figure 2 shows that focal-event mixing primarily improves event detection under non-trivial residual load. The largest gains occur for washing-machine and microwave events above 200 W. The effect is not universal: microwave recall decreases by 7.9 percentage points below 100 W, consistent with the unchanged overall microwave F1. Figure 3 separates this detection effect from power estimation. Focal-event mixing substantially reduces ON MAE for kettle and microwave across several background intervals, and also improves the high-background estimates of dishwasher and washing machine. This explains why Table VIII records better ON MAE despite a modest increase in aggregate MAE.

Figure 4 shows the distinct role of appliance-wise balancing. With a fixed task weight, the median effective state term is only 2.0% and 0.8% of the corresponding power loss for kettle and microwave. Balancing raises these ratios to 28.5% and 15.1%, respectively, while moving dishwasher and washing-machine state terms toward the power-loss scale. It therefore corrects an appliance-dependent task-scale mismatch rather than changing the event distribution. The two strategies therefore address complementary limitations: event coverage and task-scale mismatch. Their combined per-appliance outcome is shown in Fig. 5; microwave F1 changes from 0.352 ± 0.026 to 0.348 ± 0.027, so no microwave detection improvement is claimed.

VI. Conclusion

This work presented MART-NILM, a joint power-and-state model evaluated under whole-house transfer across UK-DALE and REFIT. The ablations show that its shared temporal encoder and appliance-specific decoders provide the principal architectural gains. Focal-event mixing broadens residual-background coverage, while bounded appliance-wise balancing improves the joint objective; together they increase validation macro-F1 without increasing model size. Microwave detection and fridge operation under high residual load remain the principal failure cases, motivating future work on timing-robust event representations and uncertainty-aware separation.

Appendix A
Conservative Event-Level Timing Audit

Each logged REFIT row provides a common timestamp but not the physical acquisition time of every individual meter reading. Consequently, the recorded data cannot identify

TABLE V
POWER AND ON-OFF RESULTS ON UK-DALE AND REFIT (MEAN ± STANDARD DEVIATION OVER THREE SEEDS).

Metric	Model	UK-DALE						REFIT					
		Kettle	Fridge	DW	WM	MW	Avg.	Kettle	Fridge	DW	WM	MW	Avg.
MAE (W)	UNet-NILM [5]												
	MAT-Conv [7]												
	MART-NILM	3.8 ±0.3	20.3 ±0.4	4.4 ±0.4	5.3 ±0.3	3.4 ±0.1	7.5 ±0.1	5.7 ±0.1	32.6 ±1.5	4.8 ±0.3	3.6 ±0.2	7.3 ±0.4	10.8 ±0.4
SAE (W)	UNet-NILM												
	MAT-Conv												
	MART-NILM	2.9 ±0.2	13.6 ±0.6	2.8 ±0.5	4.1 ±0.5	3.0 ±0.1	5.3 ±0.1	2.3 ±0.2	15.0 ±1.6	3.6 ±0.1	2.9 ±0.2	5.6 ±0.3	5.9 ±0.4
ON MAE (W)	UNet-NILM												
	MAT-Conv												
	MART-NILM	388.0 ±38.1	26.8 ±1.0	96.0 ±9.6	164.0 ±19.2	208.5 ±21.1	176.7 ±7.7	791.8 ±22.3	40.7 ±1.9	156.9 ±17.9	167.7 ±2.2	388.9 ±22.6	309.2 ±4.6
OFF MAE (W)	UNet-NILM												
	MAT-Conv												
	MART-NILM	1.5 ±0.2	14.7 ±0.2	1.8 ±0.3	3.1 ±0.0	2.5 ±0.0	4.7 ±0.1	1.9 ±0.0	25.1 ±1.8	2.2 ±0.2	0.9 ±0.2	6.0 ±0.3	7.2 ±0.4
Precision	UNet-NILM												
	MAT-Conv												
	MART-NILM	0.94 ±0.02	0.88 ±0.01	0.63 ±0.06	0.92 ±0.02	0.71 ±0.00	0.82 ±0.01	0.76 ±0.01	0.64 ±0.00	0.62 ±0.04	0.85 ±0.07	0.43 ±0.03	0.66 ±0.01
Recall	UNet-NILM												
	MAT-Conv												
	MART-NILM	0.99 ±0.00	0.94 ±0.01	0.99 ±0.01	0.97 ±0.02	0.91 ±0.03	0.96 ±0.00	0.81 ±0.01	0.79 ±0.02	0.77 ±0.04	0.85 ±0.09	0.76 ±0.03	0.80 ±0.03
F1	UNet-NILM												
	MAT-Conv												
	MART-NILM	0.96 ±0.01	0.91 ±0.00	0.77 ±0.04	0.94 ±0.02	0.80 ±0.01	0.88 ±0.01	0.79 ±0.00	0.71 ±0.01	0.69 ±0.01	0.85 ±0.05	0.55 ±0.03	0.72 ±0.02

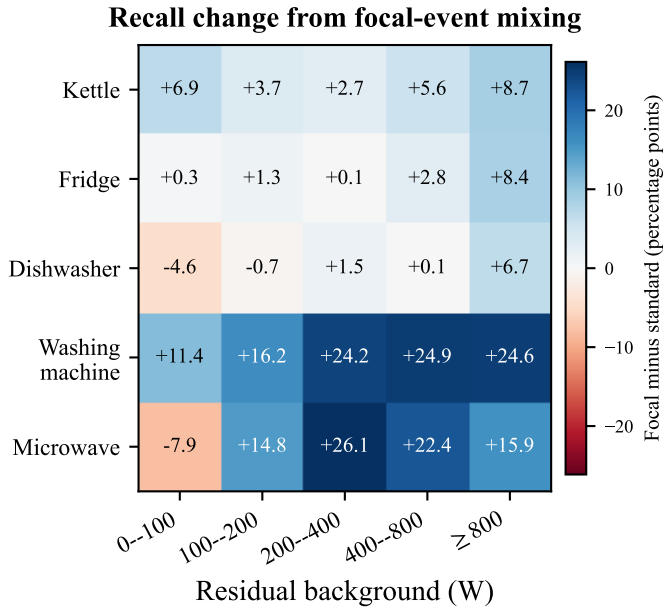


Fig. 2. Background-conditioned validation recall change produced by focal-event mixing (paired three-seed mean). Each entry is focal-event mixing minus standard training in percentage points; positive values denote improved recall.

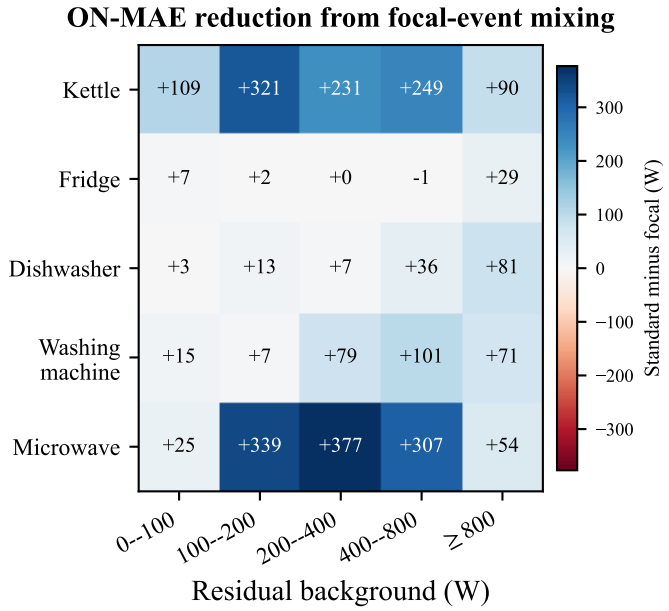


Fig. 3. Background-conditioned validation ON-MAE reduction produced by focal-event mixing (paired three-seed mean). Each entry is standard training minus focal-event mixing in watts; positive values denote lower ON MAE.

an exact, universal IAM–aggregate delay. A single nearby aggregate step is also insufficient evidence for a shift, since an unmonitored load can produce that step. We therefore use a deliberately conservative event-level audit rather than claim physical synchronization.

Let x_t and $P_{t,i}$ denote the recorded aggregate and appliance powers, respectively, and define $\Delta x_t = x_t - x_{t-1}$. Consider a complete on period $[s, e)$ of appliance i, with a rising edge $a_i^{\uparrow} = P_{s,i} - P_{s-1,i}$ and a falling edge $a_i^{\downarrow} = P_{e-1,i} - P_{e,i}$. Only events whose two appliance edges are at least $\max(40, 0.3\tau_i)$ W are considered, where τ_i is the appliance on threshold. We examine lags $\ell \in \{-2, -1, 0, 1, 2\}$, corresponding to at most ±16 s. A lag is admissible only if both aggregate edges have the expected direction and sufficient

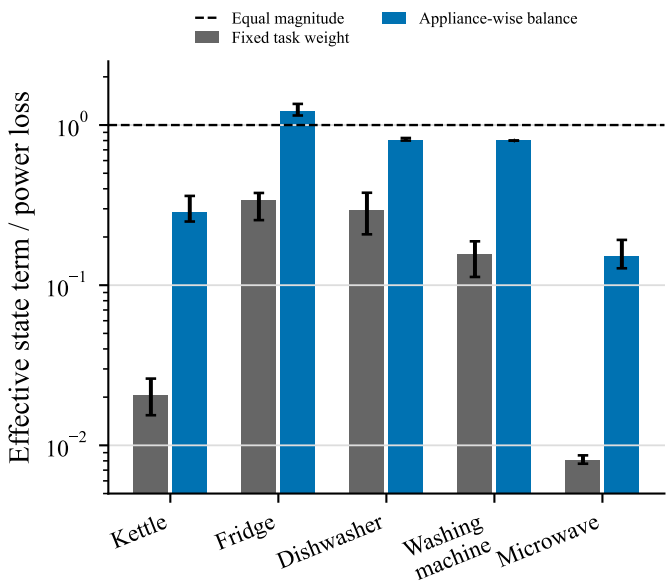


Fig. 4. Power–state scale alignment produced by appliance-wise balancing. Bars report the median effective state-term–to–power-loss ratio after epoch 20; error bars show interquartile ranges. The dashed line denotes equal numerical magnitude.

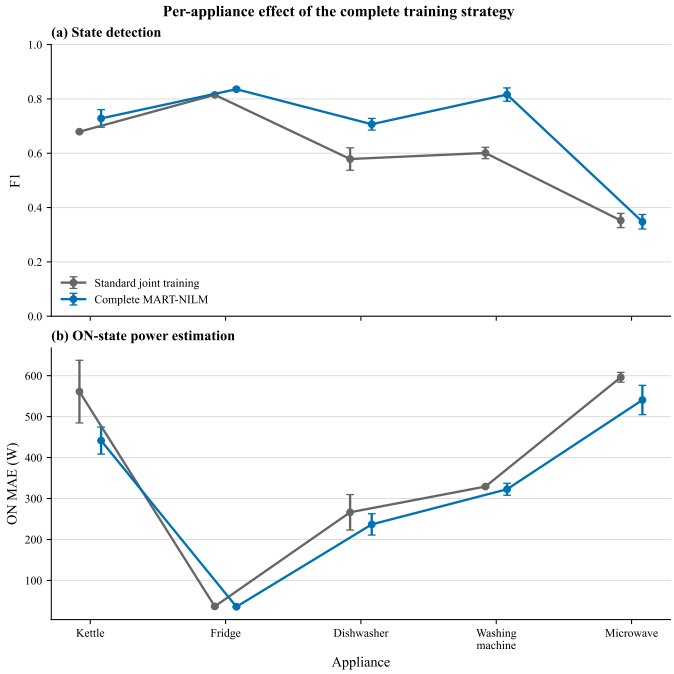


Fig. 5. Per-appliance validation effect of the complete training strategy (three-seed mean ± standard deviation). The strategy improves F1 for four appliances and reduces ON MAE for all five; microwave F1 is statistically unchanged.

TABLE VI
COMPUTATIONAL COST.

Model	Params (M)	Model size (MB)	Time (s)
UNet-NILM	5.832		
MAT-Conv	3.022		
MART-NILM	1.373	5.335	485.2

TABLE VII
ARCHITECTURE ABLATION ON THE TWO TEST HOUSES (THREE-SEED MEAN ± STANDARD DEVIATION). EVERY ROW IS ONE JOINT FIVE-APPLIANCE MODEL.

Variant	Params (M)	MAE (W) ↓	ON MAE (W) ↓	SAE (W) ↓	Macro-F1 ↑
w/o multi-scale CNN	1.372	9.22 ±0.18	241.75 ±8.76	5.66 ±0.17	0.795 ±0.001
w/o feature encoder	1.326	9.16 ±0.06	252.01 ±2.21	5.59 ±0.07	0.790 ±0.006
w/o temporal encoder	0.634	10.56 ±0.33	288.07 ±10.25	6.56 ±0.16	0.710 ±0.013
w/o appliance decoders	0.836	9.22 ±0.66	265.84 ±4.30	5.63 ±0.63	0.783 ±0.015
w/o relation attention	1.332	9.18 ±0.09	**230.84 ±14.69**	**5.55 ±0.04**	0.790 ±0.015
Complete MART-NILM	1.373	**9.12 ±0.14**	242.94 ±5.35	5.58 ±0.17	**0.796 ±0.008**

magnitude:

$$\begin{aligned} \Delta x_{s+\ell} &\geq \max(20, 0.25a_i^{\uparrow}), \\ -\Delta x_{e+\ell} &\geq \max(20, 0.25a_i^{\downarrow}). \end{aligned} \tag{25}$$

TABLE VIII
TRAINING-STRATEGY ABLATION ON THE VALIDATION HOUSES (THREE-SEED MEAN ± STANDARD DEVIATION). † DENOTES A STRATEGY PROPOSED IN THIS WORK.

Training strategy	MAE (W) ↓	ON MAE (W) ↓	SAE (W) ↓	Macro-F1 ↑
Standard joint training	21.16 ±0.39	357.81 ±19.28	15.94 ±0.35	0.605 ±0.012
+ Focal-event mixing†	21.68 ±0.14	**301.62 ±16.20**	15.60 ±0.37	0.649 ±0.016
+ Appliance-wise balancing†	20.93 ±0.07	334.86 ±10.89	15.18 ±0.32	0.661 ±0.014
Complete MART-NILM†	**20.79 ±0.18**	315.41 ±10.41	**15.02 ±0.39**	**0.687 ±0.003**

For every admissible lag, the two-edge discrepancy is

$$C_i(\ell) = \frac{1}{2}\left(\frac{|\Delta x_{s+\ell} - a_i^{\uparrow}|}{\max(a_i^{\uparrow}, 40)} + \frac{|-\Delta x_{e+\ell} - a_i^{\downarrow}|}{\max(a_i^{\downarrow}, 40)}\right). \tag{26}$$

The selected lag is $\ell^* = \arg\min_\ell C_i(\ell)$, with zero lag preferred only when scores are tied. We accept it only when $C_i(\ell^*) \leq 0.75$, its margin over the next-best admissible lag is at least 0.20, neither event boundary lies near a ≥ 40 W transition of another monitored target, and the shifted event would not overlap an existing on period. Events failing any condition are left unchanged.

For an accepted non-zero lag, the aggregate is immutable. Let b_i be the median measured power in the nearby off samples and let $q_{t,i} = \max(P_{t,i} - b_i, 0)$ within $[s, e)$. The procedure

Algorithm 1 Conservative two-edge event timing audit.

Require: 8-s aggregate x_t, appliance powers $P_{t,i}$, states $z_{t,i}$, thresholds τ_i, and $L = 2$ samples.

Ensure: Audited appliance series $\widetilde{P}_{t,i}$ and states $\widetilde{z}_{t,i}$; aggregate x_t is unchanged.

```
Set P̃_{t,i} ← P_{t,i} and z̃_{t,i} ← z_{t,i}.
for each target appliance i ∈ {1, ..., M} do
    for each complete on period [s, e) in z_{t,i} do
        Reject the event if an edge is weak, an endpoint is unavailable, or another target changes within L samples of either boundary.
        Form the admissible lag set using (25) and discard lags whose two relative edge errors exceed one.
        if the best lag satisfies the score, margin, and non-overlap tests then
            Move the excess profile and the complete state interval by ℓ* using (27).
        end if
    end for
end for
Verify Σ_t P̃_{t,i} = Σ_t P_{t,i} for every appliance and x_t is bitwise unchanged.
return x_t, P̃_{t,i}, and z̃_{t,i}.
```

moves the excess event profile as a whole:

$$\widetilde{P}_{t,i} = P_{t,i} - q_{t,i}\mathbf{1}_{[s,e)}(t) + q_{t-\ell^*,i}\mathbf{1}_{[s+\ell^*,e+\ell^*)}(t), \qquad \widetilde{x}_t = x_t. \tag{27}$$

Thus $\sum_t \widetilde{P}_{t,i} = \sum_t P_{t,i}$ for every shifted event. The associated binary on period is shifted by the same lag. This preserves the recorded input and the appliance watt-sample energy; it does *not* impose $x_t \geq \sum_i P_{t,i}$, because the measured target set is incomplete and the recorded aggregate must not be reconstructed using its labels.

The audit is a sensitivity analysis, not part of the retained training pipeline. In our cross-house ablation it slightly changed local timing errors but did not consistently improve F1 or AP across the two held-out datasets. The primary results therefore use the unshifted, mean-resampled recordings. This distinction avoids treating a plausible timing hypothesis as ground truth.

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